



**COMPENDIUM OF
INDIAN STANDARDS
ON
ELECTRICAL
TRANSFORMERS
PART 2
INSTRUMENT
TRANSFORMERS,
TRANSFORMERS FOR
SPECIAL
APPLICATIONS &
SAFETY OF
TRANSFORMERS**

**Prepared by
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TABLE OF CONTENTS

Content	Page No.
Indian Standards on Instrument Transformers	3-4
Indian Standards on Indian Standards on Welding Transformers	4-5
Indian Standards on Transformers used in Mining Applications	5
Indian Standards on Transformers used for Specific Applications	5
Indian Standards on the application guides of Instrument Transformers	5-6
Indian Standards on Safety of Transformers, Reactors, Power Supply Units and Combinations Thereof	6-7

Indian Standards on Instrument Transformers

IS 16227 (Part 1):2016-Instrument Transformers Part 1 General Requirements

This standard lays down the general requirements for newly manufactured instrument transformers including both analogue/digital output and indoor/outdoor types intended for use with electrical measuring instruments and protective devices within the frequency range of 15 Hz to 100 Hz. It also covers the requirements including service conditions, ratings, design & construction, Insulation Levels, Temperature rise, electromagnetic compatibility and tests to ensure compliance, reliability, and safety of instrument transformers.

IS 16227 (Part 2):2016- Instrument transformers: Part 2 additional requirements for current transformers

This standard specifies the additional requirements and tests for inductive current transformers used for measurement and protection in the frequency range of 15 Hz to 100 Hz. It categorizes current transformers into measuring and protective types, and further classifies protective CTs into classes P, PR, PX, PXR, TPX, TPY, and TPZ based on remanence, accuracy, and transient performance. It covers requirement with respect to design & construction, accuracy classes, composite errors, saturation characteristics, and instrument security factors, as well as standard values for rated outputs, burdens, and thermal rating.

IS 16227 (Part 3):2015- Instrument transformers: Part 3 additional requirements for inductive voltage transformers

This standard specifies the additional requirements and tests for inductive voltage transformers used with electrical measuring and protection devices in the frequency range of 15 Hz to 100 Hz. It outlines requirements for measuring and protective voltage transformers, including those with residual voltage windings, and covers requirements with respect to design & construction, accuracy classes, rated output, thermal limits, voltage factors and marking.

IS 16227 (Part 4): 2015-Instrument Transformers: Part 4 Additional requirements for combined Transformers

This standard specifies the additional requirements and tests for combined instrument transformers (devices that incorporate both a current transformer and a voltage transformer within a single enclosure—for use with electrical measuring instruments and electrical protective devices at frequencies from 15 Hz to 100 Hz. It addresses the mutual influence between current and voltage transformers, ensuring accuracy limits are maintained even under combined operation.

IS 16227 (Part 5):2015-Instrument Transformers Part 5-Additional Requirements for Capacitive Voltage Transformers

This standard lays down the additional requirements and tests for Capacitor Voltage Transformers (CVTs) connected between line and ground in systems with voltages ≥ 72.5 kV, operating at power frequencies from 15 Hz to 100 Hz. It covers requirements including design & construction, accuracy class, rated output, insulation level, marking and tests to ensure safe and reliable performance of CVT in high-voltage systems.

IS 16227 (Part 6):2018-Instrument Transformers Part 6-additional general requirements for Low-Power Instrument Transformers

This standard specifies additional general requirements and tests for Low-Power Instrument Transformers (LPITs) used for AC applications having rated frequencies from 15 Hz to 100 Hz covering

Medium Voltage, High Voltage and Extra High Voltage and also for DC applications. It applies to both low-power current transformers (LPCTs) and voltage transformers (LPVTs), with analogue or digital outputs, typically producing ≤ 1 VA. It covers requirements with respect to service conditions, accuracy class, Rated Outputs, marking and tests to ensure safe and reliable performance of Low-Power Instrument Transformers (LPITs).

IS/IEC 60044-7:1999- Instrument transformers: Part 7 electronic voltage transformers

This standard specifies the requirements and tests for electronic voltage transformers (EVTs) with analogue output, used for measurement and protection in power systems operating at 15 Hz to 100 Hz. It specifies requirements with respect to service conditions, rated primary and secondary voltages, voltage factors, burden, accuracy classes, insulation levels, short-circuit withstand capability, electromagnetic compatibility and other tests ensure the transformer's reliability and safety under both normal and transient conditions.

IS/IEC 60044-8:2002-Instrument Transformers: Part 8 Electronic Current Transformers

This standard specifies the requirements and tests for Electronic Current Transformers (ECTs) with either analogue voltage output or digital output, for use with for use with electrical measuring instruments and electrical protective devices at nominal frequencies from 15 Hz to 100 Hz. It specifies requirements with respect to service conditions, design, ratings, burden, accuracy classes, insulation levels, temperature rise limits, electromagnetic compatibility and other tests ensure the transformer's reliability and safety under both normal and transient conditions.

IS 6949:1973-Specification for summation current transformers

This standard specifies the requirements and tests for summation current transformers used exclusively for metering applications in conjunction with feeder current transformers, which may have identical or different transformation ratios. It specifies ratings (rated primary Current, rated secondary current, rated output), accuracy classes (0.2, 0.5 and 1.0), marking requirements and tests to ensure safe and reliable performance of summation current transformers under specified service conditions.

Indian Standards on Welding Transformers

IS 13339:1992-Multi-Operator type Arc Welding Transformers-Specification

This standard lays down the requirements of construction and methods of test for oil cooled three phase transformers of the flat characteristic type (substantially constant voltage) for manual multiple metal arc welding and associated current regulators. It does not apply to conventional single/double operator transformers and transformer rectifier sets for manual multiple metal arc welding. It specifies requirements with respect to ratings, duty cycles, design and construction, voltage regulation, power factor correction and tests to ensure safe and reliable operation under specified conditions.

IS 1851:1997-Single Operator Type Arc Welding Transformers-Specification

This standard lays down requirements and tests for air-cooled and oil-cooled arc welding equipment of the single operator transformer type, for manual, semi-automatic and automatic welding. It does not cover rectifier type welding equipment. It specifies requirements such as rated input voltage (240/415 V), open-circuit voltage limits (typically ≤ 80 V), duty cycle (60% for hand welding), and welding current ratings (up to 900 A), service conditions, design & construction and tests to ensure safety, reliability, and compliance under normal service conditions.

IS 4804 (Part 1):1968-Specification for Resistance Welding Equipment Part 1-Single Phase Transformers

This standard specifies the requirements and tests for single-phase transformers used in single phase transformer type AC resistance welding equipment, such as spot welders, seam welders, projection welders, and similar machines. It doesn't apply to special purpose resistance welding or resistance heating transformers and machines or equipment, Capacitor discharge or stored energy resistance welding equipment, Percussion welding equipment, Three-phase and frequency conversion or rectifier type resistance welding equipment. It covers aspects related to design, construction, materials, insulation, terminal markings, cooling requirements, and service conditions and tests to ensure performance and safety under normal and severe operating conditions.

Indian Standards on Transformers used in Mining Applications

IS 11333:1985-Specification for flameproof dry type transformers for use in mines

This standard specifies the requirements and tests for dry type flameproof three-phase transformers intended for use in underground mines and other hazardous areas. It specifies requirements including ratings, cooling methods, winding connections, design & construction, marking and tests to ensure safety and reliable performance in mining environments.

IS 2772:1982-Specification for Non-Flameproof Mining Transformers for use below ground

This standard covers requirements for single and three-phase non-flameproof, mining type, oil-immersed transformers for use below ground. It specifies requirements including kVA ratings of Single Phase and Three Phase transformers, rated voltage & frequency, winding connections and tapplings, terminal connections, performance, over voltage protection and earthing and tests to ensure safety and reliable performance in mining environments.

Indian Standards on Transformers used for Specific Applications

IS 12977:1990- Arc Furnace Transformers-Specification

This standard lays down the requirements and tests for three phase or single phase, 50 Hz, mineral-oil immersed indoor transformers together with supplementary series reactor, if any, of rating 250 kVA and above, primary voltage up to 33 kV, used for supplying electric power to direct arc melting furnaces.

IS 13956:1994-Testing Transformers-Specification

This standard covers the requirements and tests for testing transformers which are generally single phase and Oil Natural Air Natural (ONAN) cooled.

IS 5142:1969- Specification for continuously variable voltage Auto Transformers

This standard covers the requirements and tests of natural dry type and oil immersed manually operated, toroidally wound continuously variable voltage auto-transformer. Motor operated variable voltage auto-transformers and variable voltage auto-transformers having a partial variation of output voltage are not covered in this standard.

Indian Standards on the application guides of Instrument Transformers

IS 16227 (Part 100):2018- Instrument transformers: Part 100 guidance for application of current transformers in power system protection

This standard is applicable to inductive protective current transformers meeting the requirements of the IS 16227 (Part 2)/IEC 61869-2 standard. It may help relay manufacturers, current transformers (CT) manufacturers and project engineers to understand how a CT responds to simplified or standardized short circuit signals. Therefore, it supplies advanced information to comprehend the definition of

inductive current transformers as well as their requirements. It also discusses the question about the responsibilities in the design process for current transformers.

IS 16227 (Part 103):2018- Instrument Transformers part 103 The use of instrument Transformers for power quality Measurement

This standard is applicable to inductive and electronic instrument transformers with analogue or digital output for use with electrical measuring instruments for measurement and interpretation of results for power quality parameters in 50/60 Hz AC power supply systems. It aims at giving guidance in the usage of HV instrument transformers for measuring power quality parameters. The power quality parameters considered in this document are power frequency, magnitude of the supply voltage and current, flicker, supply voltage dips and swells, voltage interruptions, transient voltages, supply voltage unbalance, voltage and current harmonics and inter-harmonics, mains signalling on the supply voltage and rapid voltage changes.

IS 4146:1983- Application guide for voltage transformers (First Revision)

This guide covers application of voltage transformers for use with both electrical measuring instruments and meters (measuring voltage transformer) and electrical protective devices including broken delta voltage transformers for the application of directional earth fault protection (protective voltage transformers). Reference has also been made to the use of voltage transformer for the dual purpose of measurement and protection. This guide does not cover application of capacitor voltage transformers and phase shifting voltage transformers.

IS 4201:1983- Application guide for current transformer (First Revision)

This guide covers application of current transformers for use with both electrical measuring instruments and meters (measuring current transformers) and electrical protective devices in non-balanced protected equipment, for example, for the operation of overcurrent devices both of the non-directional and directional types and for the operation of earth fault devices with time-lag characteristics (protective current transformer). Reference has been made to the use of current transformers for the dual purpose of measurement and protection and also for the operation of those protective devices which are more precisely dependent upon the magnitude and phase relationship of voltage and current, or upon the balance of the current. This guide does not cover application of special purpose current transformer.

IS 5547:1983- Application guide for capacitor voltage transformers (First Revision)

This guide covers the application of capacitor voltage transformers for use with electrical measuring and protective devices.

Indian Standards on Safety of Transformers, Reactors, Power Supply Units and Combinations Thereof

IS/IEC 61558-1:2017- Safety of Transformers, Reactors, Power Supply Units and Combinations Thereof Part 1 General Requirements and Tests (First Revision)

This standard deals with safety aspects of transformers, reactors, power supply units and combinations thereof such as electrical, thermal and mechanical safety. This document covers independent or associated stationary or portable types of dry-type transformers thereof in the field of safety. The windings can be encapsulated or non-encapsulated. They are not forming a part of the distribution network.

IS/IEC 61558-2-4:2021- Safety of Transformers, Reactors, Power Supply Units and Combinations Thereof Part 2 Particular Requirements and Tests for Safety Isolating Transformers and Power Supply Units Incorporating Section 4 Isolating Transformers for General Applications

This standard deals with the safety of isolating transformers for general applications and power supply units incorporating isolating transformers for general applications. Transformers incorporating electronic circuits are also covered by this document. This document is applicable to stationary or portable, single-phase or polyphase, air-cooled (natural or forced) independent or associated dry- type transformers. The windings can be encapsulated or non-encapsulated. The rated supply voltage does not exceed 1 000 V AC and the rated supply frequency and the internal operating frequencies do not exceed 500 Hz. The rated output does not exceed 25 kVA for single-phase transformers and 40 kVA for polyphase transformers. The no-load output voltage or the rated output voltage does exceed 50 V AC or 120 V ripple free DC, and where applicable, does not exceed 500 V AC or 708 V ripple-free DC. The no-load output voltage and the rated output voltage may be up to 1 000 V AC or 1 415 V ripple-free DC for special applications.

IS/IEC 61558-2-6:2021-Safety of Transformers, Reactors, Power Supply Units and Combinations Thereof Part 2 Particular Requirements and Tests for Safety Isolating Transformers and Power Supply Units Incorporating Section 6 Safety Isolating Transformers for General Applications

This standard deals with the safety of safety isolating transformers for general applications and power supply units incorporating safety isolating transformers for general applications. Transformers incorporating electronic circuits are also covered by this document. This document is applicable to stationary or portable, single-phase or polyphase, air-cooled (natural or forced) independent or associated dry- type transformers. The windings can be encapsulated or non-encapsulated. The rated supply voltage does not exceed 1 000 V AC and the rated supply frequency and the internal operating frequencies do not exceed 500 Hz. The rated output does not exceed 10 kVA for single-phase transformers; 16 kVA for polyphase transformers. The no-load output voltage or the rated output voltage does not exceed 50 V AC or 120 V ripple-free DC.

IS/IEC 61558-2-7:2007-Safety of power transformers, power supplies reactors and similar products: Part 2 - 7 particular requirements and tests for transformers and power supplies for toys

This standard deals with safety aspects of transformers for toys and power supplies incorporating transformers for toys such as electrical, thermal and mechanical safety and applicable to transformers for toys and power supplies incorporating both transformers for toys and electronic circuits. It applies to stationary and portable, single-phase, air-cooled (natural or forced), transformers for toys and power supplies incorporating transformers for toys, having a rated supply voltage not exceeding 250 V AC, a rated supply and internal operating frequency not exceeding 500 Hz, a rated output not exceeding 200 VA and a rated output current not exceeding 10 A. It is applicable to dry-type transformers for toys. The windings may be encapsulated or non-encapsulated. The no-load output voltage does not exceed 33 V AC for transformers for toys and power supplies incorporating transformers for toys, and the rated output voltage does not exceed 24 V AC. for transformers and power supplies, or 33 V ripple-free DC for power supplies.